COMP4141 Slides Week 8

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Admin

Feedback

For the star system:

1. Minimally satisfactory: has a construction that can be made to work, shows some understanding of the proof, but it could have been done in more detail.
2. correct construction with a good level of proof details, only minor errors
3. Excellent answer with very detailed and correct proof.

Again stars do **not** count towards marks.

P and NP

Definition of P and NP

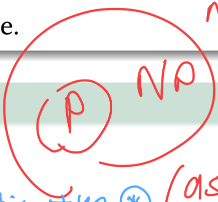
e.g. TIME($n \log n$) $\in$ P

TIME(n^k) = { Language L s.t. L can be decided by a TM in time n^k }

Definition (P)

$$\mathrm{P} = \bigcup_{k \in \mathbb{N}} \mathrm{TIME}(n^k),$$

i.e. the class of problems that can be solved in polynomial time.



Definition (NP)

$$\mathrm{NP} = \bigcup_{k \in \mathbb{N}} \mathrm{NTIME}(n^k),$$

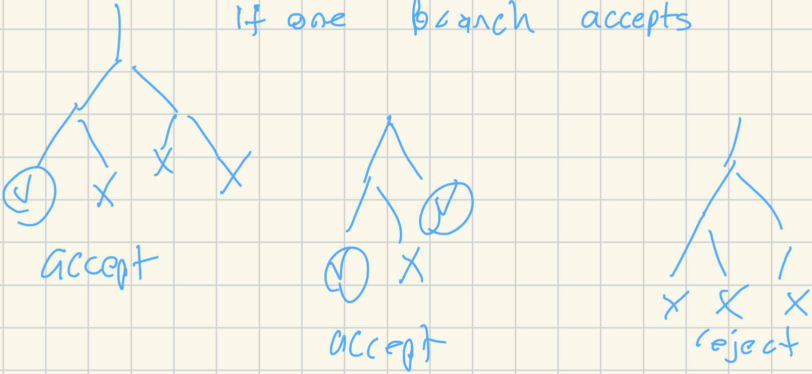
Nondeterministic TM (assuming P ≠ NP)

i.e. the class of problems that can be solved in nondeterministic polynomial time.

Run time of each branch

· We don't know if P = NP

(*) branch in the computation and a word is accepted if one branch accepts



Verifier Characterisation of NP

Verifier Definition

Certificate

A verifier for a language A is an algorithm V such that

$$A = \{ w \mid V \text{ accepts } \langle w, c \rangle \text{ for some string } c \}.$$

The language A is polynomially verifiable if it has a polynomial-time verifier (where the running time is measured in terms of $|w|$).

if V accepts ⟨w, c⟩

Theorem

A language is in NP if and only if it is polynomially verifiable.

(Yes answer)

"c (certificate) is a witness to w ∈ A"

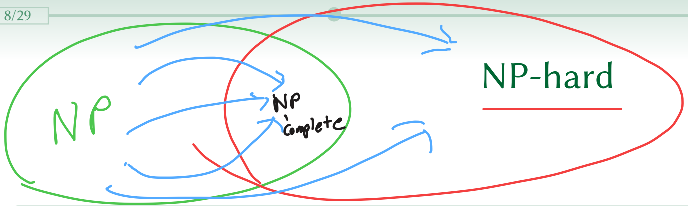
Polynomial-Time Reductions

Definition (Polynomial-Time Computable)

A function $f : \Sigma^* \to \Sigma^*$ is polynomial-time computable if there exists a polynomial-time TM M that, on input $w \in \Sigma^*$, halts with $f(w)$ on its tape.

Definition (Polynomial-Time Mapping Reduction)

Let $A \subseteq \Sigma_1^*$ and $B \subseteq \Sigma_2^*$. We say $A \leq_P B$ if there exists a polynomial-time computable function $f : \Sigma_1^* \to \Sigma_2^*$ such that $w \in A \iff f(w) \in B$.



Every other language in NP reduces to B in polynomial time

Definition

A language B is NP-hard if every $A \in \mathrm{NP}$ satisfies $A \leq_P B$.

Theorem (Proving NP-hardness)

If B is NP-hard and $B \leq_P C$, then C is NP-hard.

Typical method: reduce a known NP-hard problem P_1 to a new problem P_2. This is easier said than done!

NP-completeness

Definition

A language B is NP-complete if:

1. $B \in NP$, → can find a polynomial time verifyer,
2. B is NP-hard. → we can reduce a NP-hard problem to it.

SAT

$\neg\psi \wedge \neg\phi = \psi \vee \phi$

Definition

Let Prop $= \{x, y, \dots\}$ be Boolean variables. A Boolean formula is defined inductively by:

$$\varphi \longrightarrow p \mid \varphi \wedge \varphi \mid \neg\varphi \mid (\varphi), \qquad p \longrightarrow x \mid y \mid \cdots$$

Its truth depends on assignments to the variables.

SAT

Input: A Boolean formula φ

e.g. SAT instance could have variables x, y, z

Question: Is φ satisfiable?

There is a assignment of T\F to the boolean variables to make φ true

$\varphi = (x \wedge y) \vee (\neg y \wedge x \wedge (\neg z \vee x))$

The Grandfather of NP-Complete Problems

Theorem

SAT is NP-complete.

The proof uses a general reduction for hardness and a satisfying assignment as a certificate to show membership in NP.

Verifier then checks that φ is true under this assignment

3SAT

Definition

3-CNF

Input: A Boolean formula φ in 3-CNF

Conjunctive normal form

Question: Is φ satisfiable?

A formula is in 3-CNF if it is a conjunction of clauses, where each clause has at most three literals, e.g.

$$(x \vee y \vee z) \wedge (x \vee y \vee \neg z) \wedge (x \vee \neg y \vee z).$$

Each clause has $\vee$ over at 3 literals

"Each clause needs to be true"

W8 P1

Problem 1

Say that a formula is 4-CNF if it is in conjunctive normal form, with each clause having at most 4 literals.

$C_1 \wedge C_2 \wedge C_3 \dots \wedge C_m$

4-SAT

Input: A 4-CNF formula ϕ

C_j has at most 4 literals
over V

Question: Is ϕ satisfiable?

Show that 4-SAT is NP-Complete.

1) Show 4-SAT is in NP

- We could use the certificate as an assignment and the verifier would check that assignment is satisfied.
- Alternatively 4-SAT has subset of instance of CSAT (conjunctive normal form SAT) and since CSAT is in NP the

4-SAT is as well.

2) Want to show 4-SAT is NP-hard

We will show 3-SAT $\leq_p$ 4-SAT

boolean formula

word

Given a instance φ to 3-SAT

φ is also also an instance of 4-SAT

but also since 4-SAT $\geq$ 3-SAT ... (?)

we can use $f(\varphi) = \varphi$

so that by 1

$\varphi \in 3\text{-}SAT \iff f(\varphi) = \varphi \in 4\text{-}SAT.$

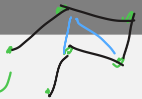
This reduction is polynomial time (idenity) so 4-SAT is NP-hard and NP-complete since it is also in NP.

since 3-SAT is NP-hard

W8 P2

Problem 2

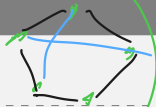
Consider the following decision problems:



Hamiltonian Path

Input: An undirected graph G

Question: Is there a path in G that visits every vertex exactly once?



Hamiltonian Cycle

Input: An undirected graph G

Question: Is there a cycle in G that contains every vertex exactly once?

(a) Show that Hamiltonian Path $\leq_P$ Hamiltonian Cycle.

Idea from G (graph input to HamPath)

for G' that has a HamCycle

iff G has a Ham Path

Can't just add a new vertex between the start and end of the Ham path since we don't know if there is one.

We will take graph $G=(V,E)$ and add new vertex v^* (not in V)

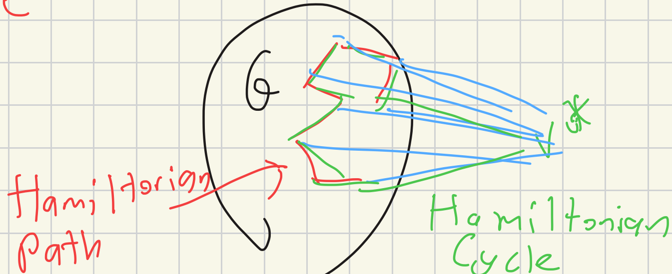
have $V' = V \cup \{v^*\}$

$\begin{pmatrix} n := |V| \\ m := |E| \end{pmatrix}$ $E' = E \cup \bigcup_{u \in V} (u, v^*)$

First of all we add 1 new vertex and n new edges which is polynomial time in our input (reduction is polynomial time)

($\Leftarrow$) If G has a Ham Path $v_1 v_2 \ldots v_n$ then G'

then $v^* v_1 v_2 \ldots v_n v^*$ is Ham Cycle

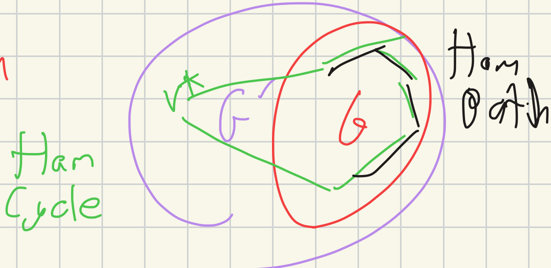


($\Leftarrow$) If G' has Ham Cycle

$v^* v_1 v_2 \dots v_n v^*$

then G has Ham path

of $v_1 v_2 \dots v_n$



Problem 2

Consider the following decision problems:

HAMILTONIAN PATH

Input: An undirected graph G

Question: Is there a path in G that visits every vertex exactly once?

HAMILTONIAN CYCLE

Input: An undirected graph G

Question: Is there a cycle in G that contains every vertex exactly once?

(b) Show that both problems are in **NP**.

We showed in (a)

Ham Path $\le_p$ Ham Cycle

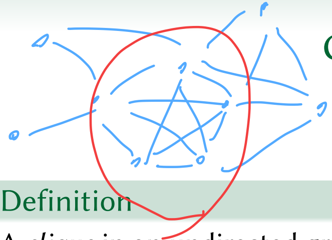
So it suffices to show that Ham Cycle is in NP

($A \le_p B$ and B is NP then so is A $A \Rightarrow B$)

To show Ham Cycle is in NP:

- Use certificate as sequence of vertices $v_1 v_2 \dots v_k$
- The verifier will check every vertex in V apears exactly once and all adjacent pairs (including endpoints)

W8 P3



Cliques and Vertex Covers

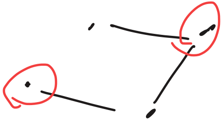
G has clique of size 5

Definition

A *clique* in an undirected graph is a fully connected subgraph. A k-clique is a clique with k vertices.

Vertex Cover

A *vertex cover* in an undirected graph $G = (V, E)$ is a set $\mathcal{V}$ such that for every edge $(u, v) \in E$, at least one of u or v belongs to $\mathcal{V}$.



Vertex Cover

Problem 3

Consider the following problems:

Clique

Input: An undirected graph G and an integer k

Question: Does G have a clique of size k?

Vertex Cover

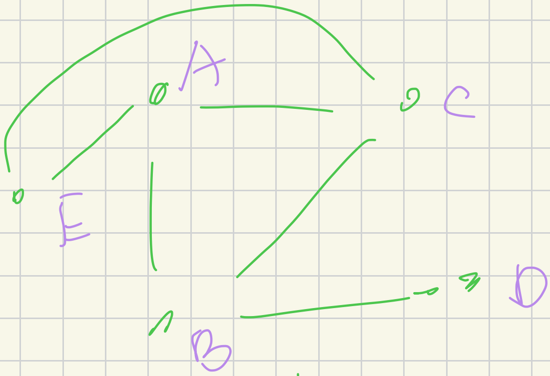
Input: An undirected graph G and an integer k

Question: Does G have a vertex cover of size k?

Show that Clique is **NP**-complete by giving a reduction from Vertex Cover.

in NP by certificate of the proposed verticies in the clique

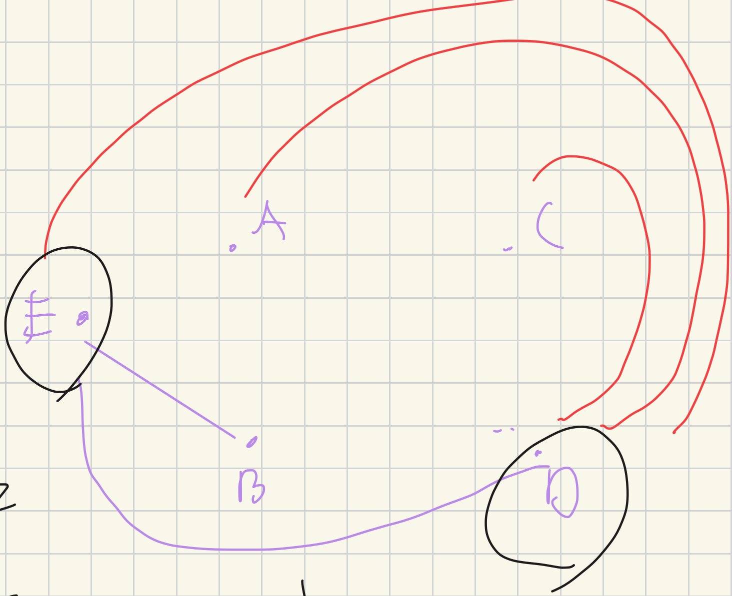
G^c



G has a clique of size 3

$\{A, B, C\}$

G
(complement graph)



Vertex cover of size 5-3=2

{D, E} make a vertex cover

- We will take input $\langle G, k \rangle$ to Vertex cover and output $\langle G^c, n-k \rangle$
- Can be computed in $O(n^2)$ time
- Want to show G has a vertex cover of size k iff G^c has clique of size $n-k$

$|S| = k$, S vertex cover,

$\forall (u,v) \in E$, $u \in S$ or $v \in S$

$\Leftrightarrow$ $|S^c| = n-k$

$\nexists (u,v) \in E$ such that $u \notin S$, $v \notin S$

$\Leftrightarrow$ $|S^c| = n-k$ $(u \in S^c, v \in S^c)$

$\forall u, v \in S^c$, $(u,v) \in E$

$((u,v) \in E^c)$

$\Leftrightarrow$ There is a clique of size $n-k$

W8 P4

Problem 4

Consider the following problem:

3-Col

Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Suppose we knew that 3-Col is **NP**-complete. Using this fact, give a new proof by reduction that SAT is **NP**-complete.

we want to show 3-Col $\leq$ SAT

Given graph G in 3-col
$G = (V, E)$

we will create boolean variable

$$x_{v,c}$$

"If $x_{v,c}$ is true then $v \in V$ has colour c"

- Need no vertex has more than one colour:

$3\binom{3}{2} \cdot |V|$

$$\bigwedge_{v \in V, c_1 \neq c_2} \neg x_{v,c_1} \vee \neg x_{v,c_2}$$

- Need no adjacent colours

$3|E|$

$$\bigwedge_{(u,v) \in E, c} \neg x_{v,c} \vee \neg x_{u,c}$$

- Every vertex need a colour

$$|V| \quad \bigwedge_{v \in V} (x_{v,g} \vee x_{v,b} \vee x_{v,r})$$

So polynomial number of clauses,
so reduction is polynomial time

$f(G)$ is satisfiable iff
G has a 3 colouring

(Elaboration: set $x_{v,c}$ if v has color c in forward and color $x_{v,c}$ if v has color c in the reverse)

W8 P5

Problem 5

Consider the following problem:

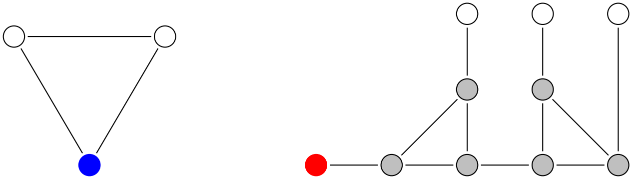
3-COL

Input: An undirected graph G

Question: Is there a colouring of the vertices of G using at most 3 colours such that no edge has both endpoints with the same colour?

Problem 5

(a) What can be said about the colouring of the white vertices in the following (partially coloured) graphs?



Problem 5

(b) Show that 3-Col is **NP**-complete, assuming that SAT is **NP**-complete.

W8 P6

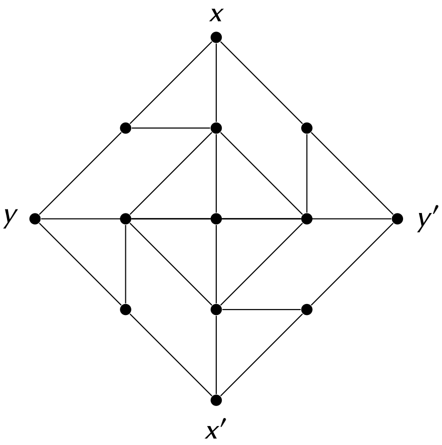
Problem 6

Show that if $\mathbf{P} = \mathbf{NP}$, then every language $A \in \mathbf{P}$ except $A = \emptyset$ and $A = \Sigma^*$, is $\mathbf{NP}$-complete.

Bonus

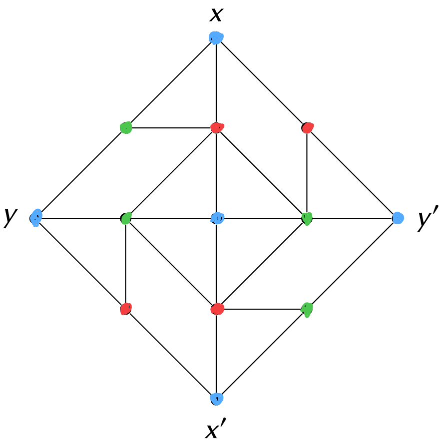
Bonus

Consider the following graph:



Bonus

Consider the following graph:



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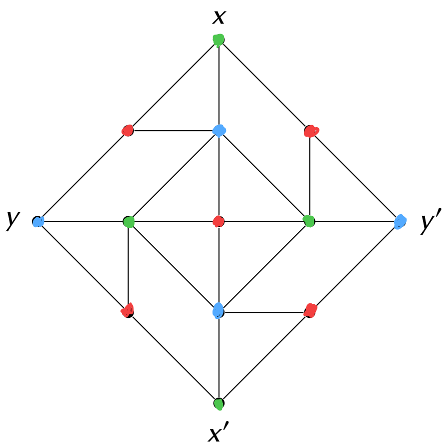
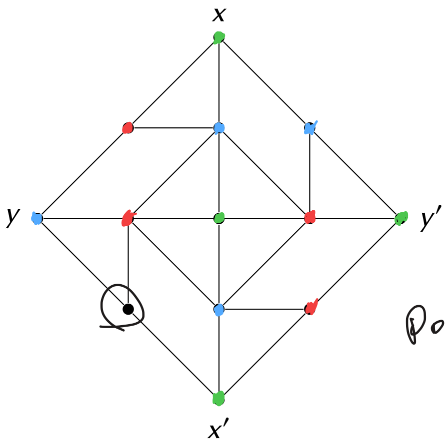


Bonus

Consider the following graph:



Does not work

Bonus

(a) Give a valid 3-colouring of this graph where x and y have different colours.

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(b) Give a valid 3-colouring of this graph where x and y have the same colour.

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(a) Give a valid 3-colouring of this graph where x and y have different colours.

(b) Give a valid 3-colouring of this graph where x and y have the same colour.

(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

Bonus

(a) Give a valid 3-colouring of this graph where x and y have different colours.

(b) Give a valid 3-colouring of this graph where x and y have the same colour.

(c) What can be said about *any* valid 3-colouring of this graph? Justify your answer.

(d) Consider the following decision problem:

Planar 3-Col

Input: A planar graph G

Question: Does G have a valid 3-colouring?

Show that Planar 3-Col is NP-complete.
Hint: Use the above graph to "uncross" edges.

Bonus

(e) Prove that there is no planar graph G, with x, y, x', y' on an outer face (in that order), that satisfies the conditions indicated in (a), (b) and (c), but with a 4-colouring rather than a 3-colouring.